

ZHANG PING, China

WMO: 589260

Lat: 25.2980N

Lon: 117.4137E

Elev: 203

StdP: 98.91

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.1	(c) 4.0	(d) -4.6	(e) 2.6	(f) 9.6	(g) -1.6	(h) 3.4	(i) 9.6	(j) 5.2	(k) 16.1	(l) 4.3	(m) 14.5	(n) 0.8	(o) 20	(p) 0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 36.2	(d) 25.0	(e) 35.2	(f) 24.9	(g) 34.1	(h) 24.9	(i) 26.6	(j) 32.3	(k) 26.1	(l) 31.6	(m) 25.8	(n) 30.9	(o) 2.2	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.3	(b) 21.0	(c) 27.7	(d) 25.0	(e) 20.6	(f) 27.4	(g) 24.6	(h) 20.1	(i) 27.3	(j) 84.5	(k) 32.6	(l) 82.6	(m) 31.7	(n) 81.1	(o) 31.1	(p) 29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.0	4.1	3.3	-0.6	38.0	1.7	0.8	-1.8	38.6	-2.8	39.1	-3.7	39.5	-4.9	40.1
			-1.3	27.2	1.9	0.8	-2.6	27.8	-3.7	28.2	-4.7	28.6	-6.1	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	12.8	14.7	17.6	21.7	24.8	27.3	29.0	28.7	27.0	23.3	18.7	13.9		
	DBStd	6.45	3.70	4.42	3.86	3.38	2.66	2.13	1.62	1.73	2.28	2.80	3.36	3.53		
	HDD10.0	31	14	9	1	0	0	0	0	0	0	0	0	7		
	HDD18.3	542	173	115	62	11	1	0	0	0	0	2	34	144		
	CDD10.0	4284	102	142	236	350	458	518	588	581	509	412	262	127		
	CDD18.3	1753	3	14	38	111	201	268	329	322	259	156	46	6		
	CDH23.3	16025	11	90	227	723	1554	2563	3734	3386	2403	1098	221	15		
(13)	CDH26.7	6547	0	14	44	227	570	1025	1778	1527	991	331	39	1		
(14)	Wind	WSAvg	1.3	1.1	1.2	1.3	1.3	1.3	1.5	1.4	1.3	1.2	1.1	1.1		
(15) Precipitation	PrecAvg	1592	65	96	159	174	227	249	147	212	118	48	48	49		
	PrecMax	2424	269	273	389	360	407	496	266	436	267	193	197	165		
	PrecMin	1051	3	4	69	46	77	98	25	78	12	0	0	0		
	PrecStd	297	56	67	72	80	97	91	64	84	57	50	55	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	28.7	30.0	32.7	35.1	36.2	37.5	37.2	35.9	33.0	30.0	25.2		
		MCWB	17.4	19.4	20.8	22.1	24.1	25.3	24.8	25.1	24.7	23.4	21.6	18.8		
	2%	DB	22.7	26.0	27.9	30.9	33.1	34.7	36.3	36.0	34.4	31.3	27.6	22.7		
		MCWB	16.4	18.2	19.3	21.7	22.9	24.9	24.9	25.1	24.6	22.5	19.9	17.1		
	5%	DB	20.7	23.9	26.0	29.4	31.6	33.4	35.3	34.8	33.2	29.9	25.8	20.9		
		MCWB	15.2	17.2	18.6	21.3	22.9	24.7	24.9	25.2	24.2	21.5	18.9	16.0		
	10%	DB	18.7	21.5	23.9	27.6	30.0	32.0	34.0	33.5	31.8	28.5	24.2	19.3		
		MCWB	14.0	16.1	17.9	20.4	22.7	24.6	25.1	25.2	23.9	20.9	18.2	15.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	20.4	22.0	23.9	26.2	27.0	27.1	27.0	26.5	25.0	22.7	20.5		
		MCDB	22.9	26.8	27.8	29.5	31.3	32.5	33.0	33.6	32.3	30.3	28.1	22.6		
	2%	WB	17.4	19.1	20.8	23.1	25.2	26.3	26.4	26.4	25.8	24.2	21.6	18.6		
		MCDB	21.1	24.1	26.0	28.3	29.9	31.2	32.3	32.5	31.1	28.3	25.6	21.4		
	5%	WB	16.3	18.0	19.9	22.4	24.6	25.9	26.0	26.0	25.3	23.5	20.6	17.3		
		MCDB	19.6	22.3	24.0	27.2	28.9	30.2	31.7	31.7	30.0	27.3	23.8	19.9		
	10%	WB	15.1	16.9	18.9	21.7	24.0	25.5	25.7	25.6	24.8	22.6	19.7	16.0		
		MCDB	18.0	20.6	22.7	25.9	27.7	29.5	31.0	30.7	29.1	26.3	22.6	18.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.4	8.5	8.5	8.2	7.8	9.4	9.0	8.9	9.7	9.6	9.3		
		MCDBR	11.3	12.3	12.3	11.7	11.7	10.7	11.8	11.3	10.8	11.5	11.8	11.1		
		MCWBR	5.3	5.2	4.8	3.9	3.3	2.6	2.5	2.4	2.3	3.2	4.4	5.3		
	5% WB	MCDBR	9.2	10.4	9.8	9.6	8.6	8.3	9.6	9.6	9.2	8.7	8.4	7.8		
		MCWBR	4.5	4.6	4.3	3.6	3.0	2.8	2.7	2.7	2.5	2.8	3.5	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.611	0.685	0.799	0.791	0.666	0.614	0.582	0.655	0.669	0.651	0.581	0.565		
	taud		1.672	1.574	1.417	1.472	1.726	1.849	1.928	1.738	1.692	1.676	1.789	1.781		
	Ebn at Noon		631	621	580	603	684	716	738	685	659	638	653	649		
	Edh at Noon		219	259	317	307	237	208	192	232	237	230	193	189		
(44) All-Sky Solar Radiation	RadAvg		2.71	2.84	3.15	3.57	3.86	4.15	5.35	4.83	4.35	3.89	3.06	2.87		
	RadStd		0.46	0.53	0.47	0.56	0.63	0.50	0.48	0.47	0.40	0.61	0.39	0.47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.44	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.31	N/A	N/A	+0.43	N/A	N/A	N/A	N/A	+98	+88

Nomenclature: See separate page